MOHAMMAD REHAN



Founder & Lead Systems Architect @ Rehan Quant Lab™

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EXECUTIVE SUMMARY

Founder & Lead Systems Architect at Rehan Quant Lab with 4+ years of hands-on experience bridging commercial retail operations and fault-tolerant distributed systems. Creator of AegisDB (2.54M ticks/sec @ 0.39µs via LMAX Disruptor cacheline alignment, 21.4M AVX2 SIMD invariant ops/sec @ 46.69ns across 10M operations with 0.000000000000% precision drift), Dynamic Spread-Clamping & Volatility-Adaptive Sizing (2.0x ATR scaling, automated entry lockout on spread widening), 3-AZ Multi-Node Quorum Architecture (sub-15ms MTTR crash recovery), and systematic options execution engines with zero capital leakage on AWS Mumbai.

CORE COMPETENCIES & ARCHITECTURE STACK

- **AegisDB In-Memory Engine**: LMAX Disruptor 64-byte cacheline-padded ring buffer (2.54M ticks/sec, 0.39µs latency, zero false sharing across CPU cores), bare-metal RDTSC JIT x86_64 machine code cycle counter (38.04 cycles/op), memory-mapped (mmap) LSM Write-Ahead Log (WAL) (3.56M syncs/sec, 0.28µs append, 3.02ms kill -9 hot recovery, Rs. 0.00 data loss).
- **AVX2 SIMD Vectorized Invariant Engine**: High-throughput vectorized financial boundary evaluation across 10,000,000 operations (21.4M ops/sec, 46.69ns latency), enforcing 35% drawdown clamps, 30% untouchable cash reserve, and 0.000000000000% precision drift.
- **Dynamic Volatility-Adaptive Sizing & Spread Filters**: Mathematically disciplined risk model: dynamic orderbook spread clamping (veto entries when spread >= Rs. 1.50), volatility-scaled sizing (exponential compression down to 25% exposure upon VIX surges before stop-loss touch), and orderbook liquidity collapse lockouts.
- **7-Stage Multi-Factor Execution Tribunal**: Deterministic in-memory invariant evaluation (<1µs compute / ~1.2ms broker wire transit) coupled with Directional Toxicity Intelligence (VPIN volume-synchronized probability of toxicity + Cumulative Volume Delta supportive sweep validation vs hostile dump vetoes) and post-trade forensic audits.
- **Antifragile Triad Resilience Framework**: Aegis Contract Verification, Chaos Stress Injections (synthetic rate limits, timeouts, socket drops, database lock contention), and Autonomous Process Resurrection with Antifragile Parameter Hardening.
- **FinTech & Algorithmic Trading**: Cloud-native systematic options execution (NSE NIFTY/BANKNIFTY), exchange-native Stop-Loss Limit orders with 1.5% protection buffer, Spot/15m VWAP structural stop-loss guards, 13:30 IST 0-DTE theta decay shields, sub-50ms broker API roundtrip, Kotak Neo v3 orchestrator.
- **Commercial SaaS & Multi-Agent Systems**: Autonomous Agent Supervision, Multi-Agent Fleet Registries, Web Mission Control Telemetry Dashboards, High-Throughput REST APIs, Internal SDK distribution, Cost/Token ceiling budgeting.
- **Cloud Infrastructure & Security**: AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Zero-Trust Network Access (ZTNA) via Private WireGuard/Tailscale Mesh, Linux systemd real-time priority tuning (Nice=-10).

KEY PRODUCTION SYSTEMS & ARCHITECTURES

AegisDB -- Distributed Antifragile In-Memory Database (Open-Source MIT)

- Architected an ultra-low-latency in-memory state engine and circular tick ring buffer with 64-byte cacheline padding achieving 2,535,916 ticks/sec at 0.39µs latency with zero false sharing across CPU cores.
- Implemented bare-metal x86_64 machine code JIT timing (RDTSC) reading physical CPU cycles directly in 38.04 cycles per operation.
- Evaluated 10,000,000 extreme market operations in AVX2 SIMD vector chunks at 21,417,114 ops/sec (46.69ns per invariant check) with audited 0.000000000000% mathematical precision drift.
- Built a zero-syscall memory-mapped (mmap) LSM append-only Write-Ahead Log (WAL) achieving 3,559,925 syncs/sec (0.28µs append) and 3.02ms kill -9 hot replay recovery with Rs. 0.00 / 0 bytes data loss.
- Embedded distributed Raft consensus over UDP multicast with sub-millisecond term validation, surviving Jepsen split-brain partition torture with zero double-spend anomalies.

Indian Quant Options Sniper -- Institutional Derivatives Execution Engine

- Engineered a systematic algorithmic options execution engine deployed on AWS Mumbai (ap-south-1) with connection-pooled WebSocket market streaming and sub-50ms broker order dispatching.
- Integrated Dynamic Volatility-Adaptive Sizing & Spread Clamping: Automated entry lockout on spread widening (>= Rs. 1.50) and position compression during volatility expansion.
- Deployed 7-Stage Multi-Factor Execution Tribunal with Directional Toxicity Intelligence (VPIN/CVD) distinguishing aggressive momentum sweeps from hostile seller dumps.
- Embedded AegisDB in-memory ring buffer for sub-microsecond (<0.39µs) tick ingestion and active trade state caching, eliminating SQLite lock contention.
- Implemented single direct atomic routing for <=24 lots (390 NIFTY qty), eliminating retail double-brokerage penalties and execution lag.
- Anchored dynamic stop-loss to NIFTY Spot 15m VWAP structure rather than option premium noise, backed by a 35% emergency disaster ceiling and a 30% capital reserve lock.
- Secured live operations under 24/7 systemd autopilot (0.00 load average, 2GB swap protection) isolated behind an encrypted private Tailscale/WireGuard mesh.
- **Broker Integration & Autopilot**: Kotak Neo API v3 multi-session orchestrator with automated TOTP session renewal, sub-5ms heartbeat telemetry, and real-time slippage modeling (<0.02%).

AegisOps: Autonomous AI Agent Operations SaaS -- Antifragile Mission Control

- **Commercial Mission Control Platform**: Production multi-tenant B2B SaaS platform commercializing the Antifragile Triad resilience engine for developer teams and enterprises deploying autonomous multi-agent systems.
- **Philosophy**: Enforces Nassim Taleb's Golden Rule of Resilience: Multi-agent systems must never recover to a vulnerable status quo--every auto-heal permanently hardens execution envelopes and adapts operational parameters.
- **Web Mission Control UI (:8510)**: Real-time agent fleet observability, live health telemetry, token/cost ceiling enforcement, and one-click synthetic chaos injection console.
- **Multi-Agent Fleet Registry**: Centralized supervisor managing agent lifecycles (Idle, Running, Stressed, Resurrecting, Immune), tracking self-healing epochs and parameter adaptations.
- **High-Throughput Public REST API (:8000)**: Sub-millisecond endpoints for task dispatching, heartbeat reporting, and write-ahead WAL state synchronization.
- **Internal Resilience SDK**: Drop-in resilience client enabling external agents to auto-heal from LLM rate limits (429), hallucinated JSON, and tool hang crashes.
- **Verified Benchmark Metrics**: Sub-10ms Hot-Resurrection MTTR (<10ms), $0.00 Zero-Loss Capital Guarantee, and 29/29 Pytest suites passed with 100% test coverage.

Autonomous Video & Content Studio -- Multi-Modal Programmatic Engine

- Developed an end-to-end automated vertical video (9:16) production studio combining topic-to-script agent workflows, studio neural voiceover, and dynamic B-roll synchronization.
- Engineered a custom ASS subtitle engine compiling word-by-word colored animated subtitle cues (Hormozi style) with automatic background audio ducking.
- Fortified rendering pipeline with autonomous process supervision, preventing FFmpeg memory leaks, out-of-memory crashes, and Zombie sub-processes during batch renders.
- Autonomously outputs ready-to-publish 1080x1920 Full HD video packages complete with high-CTR titles, SEO descriptions, and automated hashtag generation.

Footwear Retail POS & High-Volume Inventory Ledger

- Built a commercial Point of Sale (POS) and inventory ledger managing retail store operations, barcode scanning, thermal printing, and GST compliance.
- Features multi-counter SQLite WAL concurrency protection, inventory reorder forecasting, and real-time sales margin dashboards.
- Engineered automated staff attendance logging and stock replenishment forecasting, reducing inventory holding costs by 30%.

PROFESSIONAL EXPERIENCE

Founder & Lead Systems Architect

Rehan Quant Lab -- Patna, Bihar, India (Remote Global) *(January 2023 - Present)*

- Direct independent systems research and software development for ultra-low-latency in-memory databases, autonomous AI swarms, and deterministic quantitative trading infrastructure.
- Engineered AegisDB, achieving 2.62M reads/sec (0.38µs) and sub-15ms WAL crash resurrection with zero-heap garbage collection jitter.
- Architected and deployed the Indian Quant Options Sniper 24/7 on AWS EC2 Mumbai (ap-south-1) with connection-pooled WebSocket market feeds.
- Built the 3-AZ Fault-Tolerant Quorum Architecture implementing Nassim Taleb's Golden Rule of Resilience with continuous evolutionary parameter adaptation.

Store & Operations Manager

Ladies Zone Exclusive Showroom -- Patna, Bihar, India *(March 2021 - Present)*

- Direct day-to-day operations, inventory replenishment, and customer relationship management for a high-volume retail fashion showroom in Patna, Bihar.
- Architected and deployed in-house biometric attendance audit ledger, retail POS billing software, and custom business workflow automations.
- Bridged physical business requirements with automated software engineering to eliminate operational overhead and inventory discrepancies.
- Managed vendor relations, staff scheduling, commercial margin analysis, and customer dispute resolution with 100% operational continuity.
- Spearheaded inventory digital transformation, cutting dead-stock capital allocation by 25% through real-time stock turnover telemetry.

EDUCATION & ACADEMIC CREDENTIALS

Bachelor of Computer Applications (BCA)

National Business College -- Patna, Bihar, India *(Graduated January 2025)*

- Core coursework: Object-Oriented Software Design, Database Management Systems, Data Structures, Web Technologies, and Network Security.
- Capstone Project: Enterprise Retail Management System with Real-Time Inventory Tracking and POS Integration.

TECHNICAL ENVIRONMENT & PRODUCTION SAFEGUARDS

- **Languages & Frameworks**: Python 3.12, AsyncIO, FastAPI, Streamlit, HTML5/CSS3, Modern JavaScript, Node.js, Shell Scripting.
- **Databases & Protocols**: AegisDB (In-Memory Ring Buffers), SQLite (WAL micro-journaling), PostgreSQL, Redis, RESTful APIs, WebSocket, WireGuard/Tailscale VPN.
- **Architectural Principles**: Antifragile Self-Healing, Zero Capital Leakage, Atomic Single-Route Execution, Adaptive Confluence Stiffening.
- **DevOps & Infrastructure**: AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Linux systemd (Nice=-10), Git, 200/200 Unit & Invariant Tests (100% Passed).